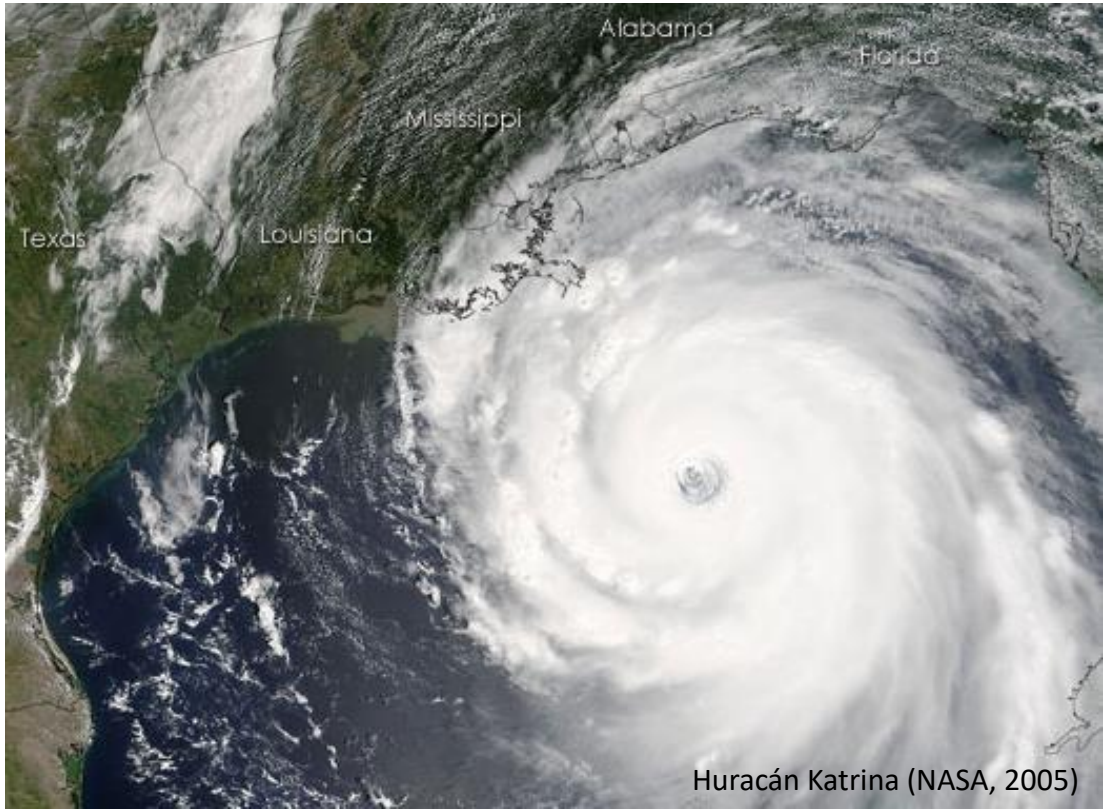
UNIVERSIDAD ADOLFO IBÁÑEZ
Facultad de Artes Liberales
Departamento de Ciencias

CURSO: CORE CIENCIAS
Cambio Global

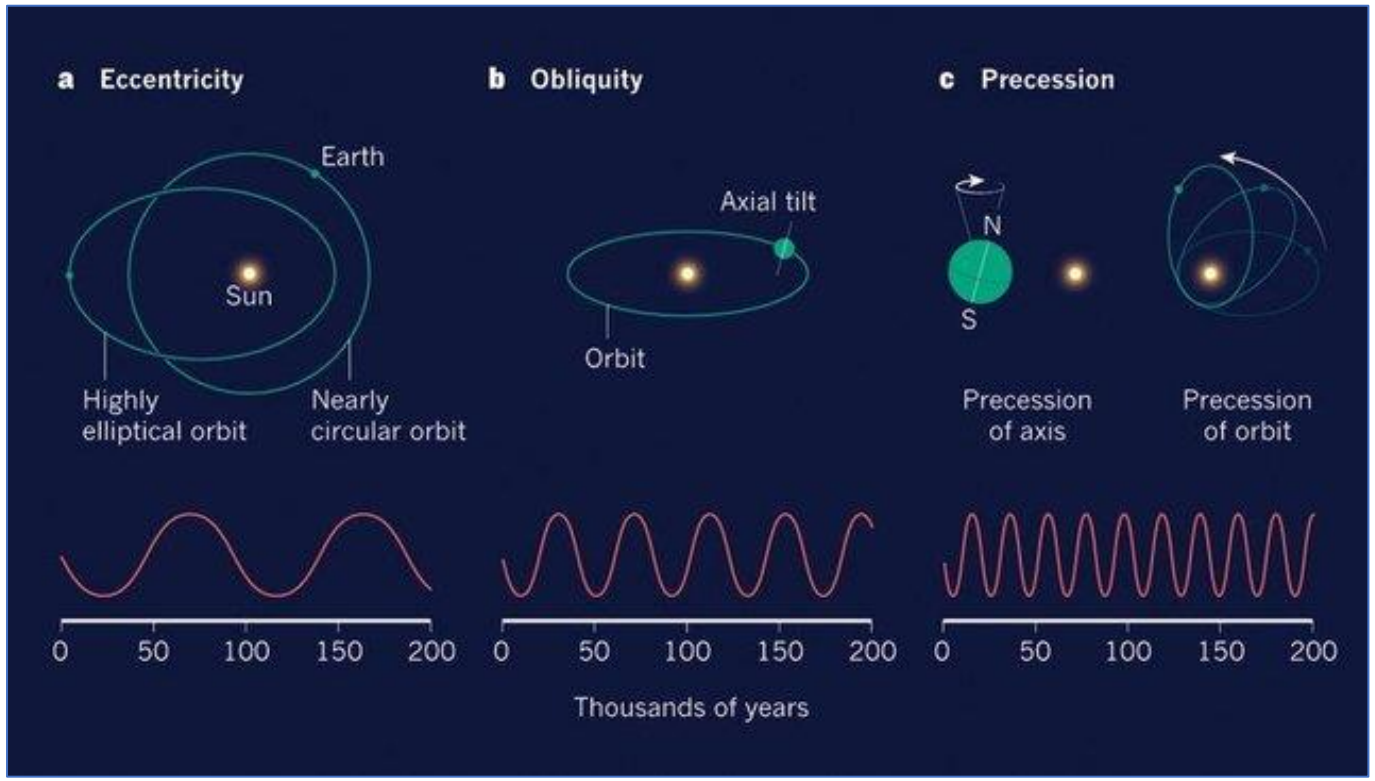


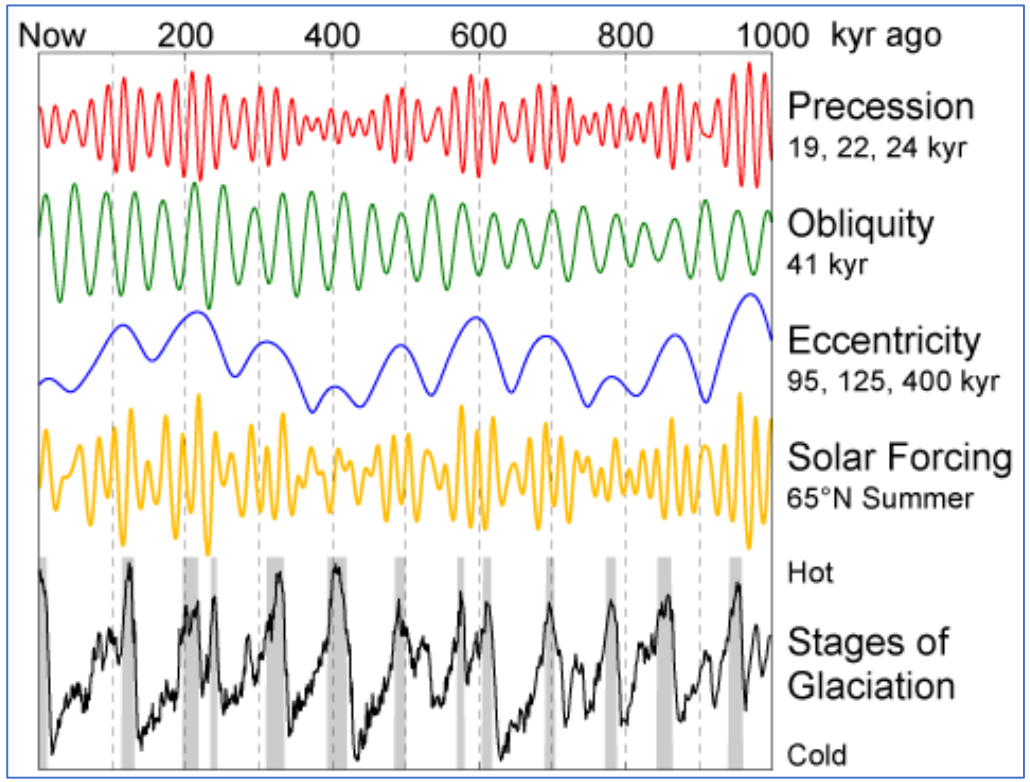
Huracán Katrina (NASA, 2005)

Prof. Ignacio Fernández
Ignacio.fernandez.c@uai.cl

Ciclos Naturales

La tierra posee **ciclos naturales** que producen cambios en la cantidad y distribución de la energía que recibe del sol, generando **modificaciones en los patrones climáticos.**

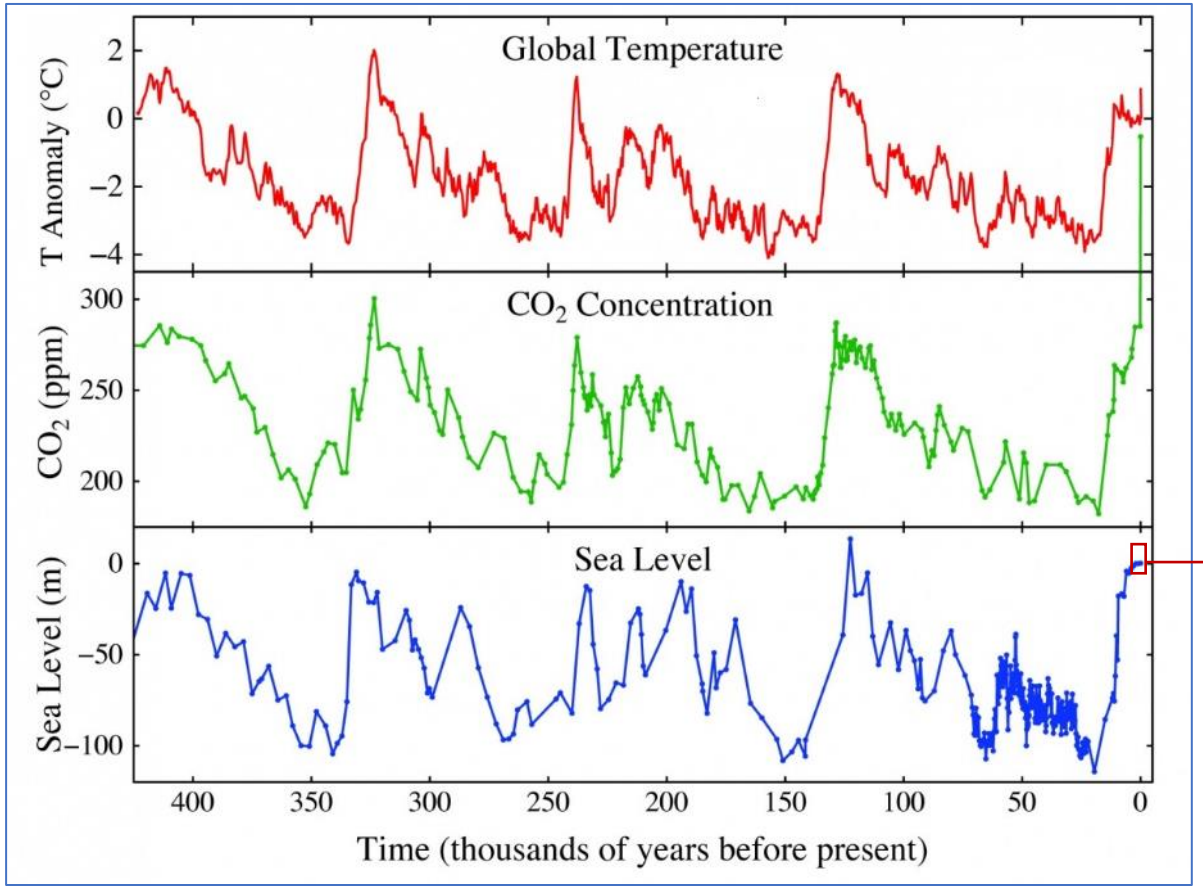


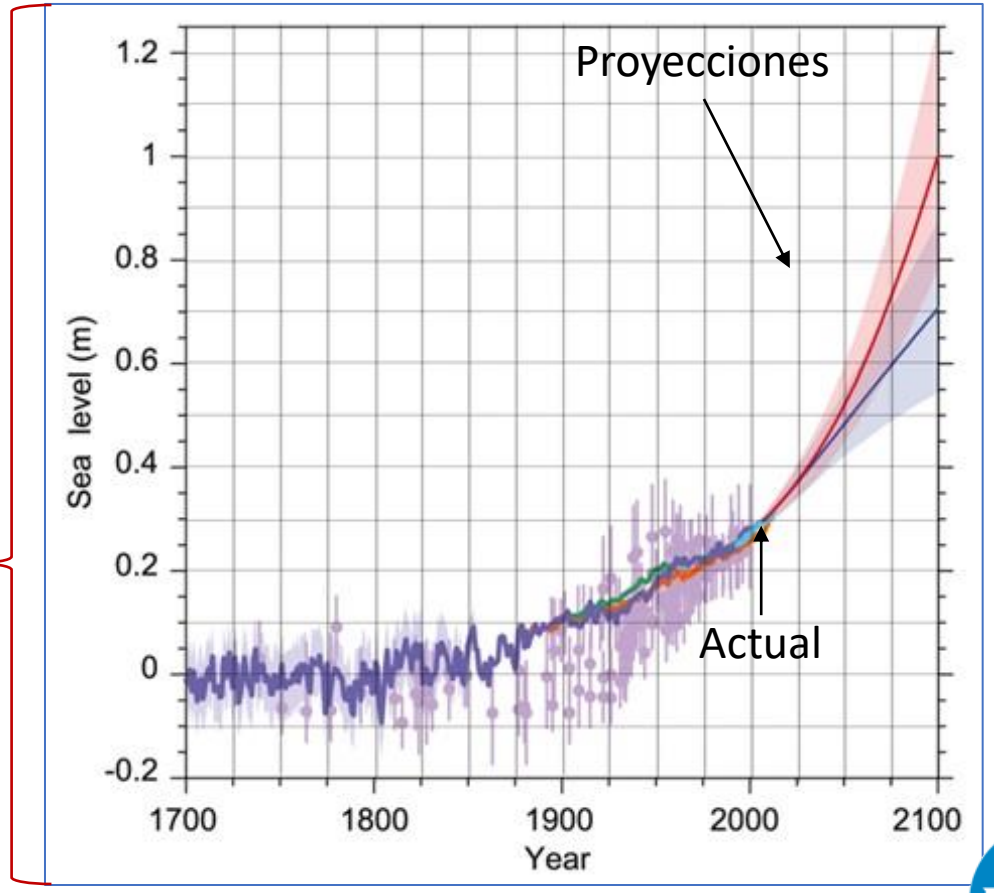


http://what-when-how.com/global-warming/milankovitch-cycles-global-warming/

Cambio climático

Actualmente el **planeta está experimentando una serie de cambios** a nivel global. Uno de los más identificables es el cambio climático.

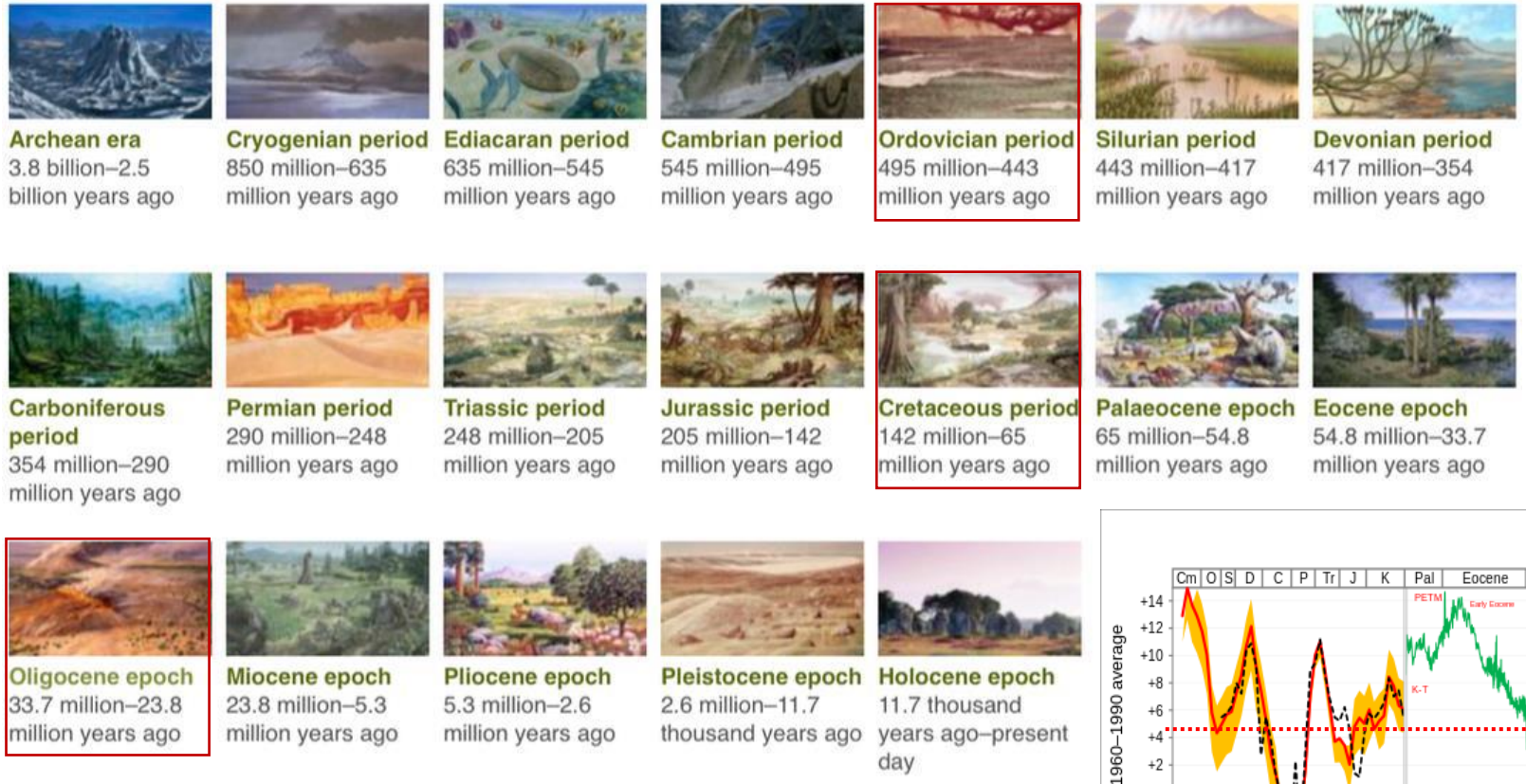




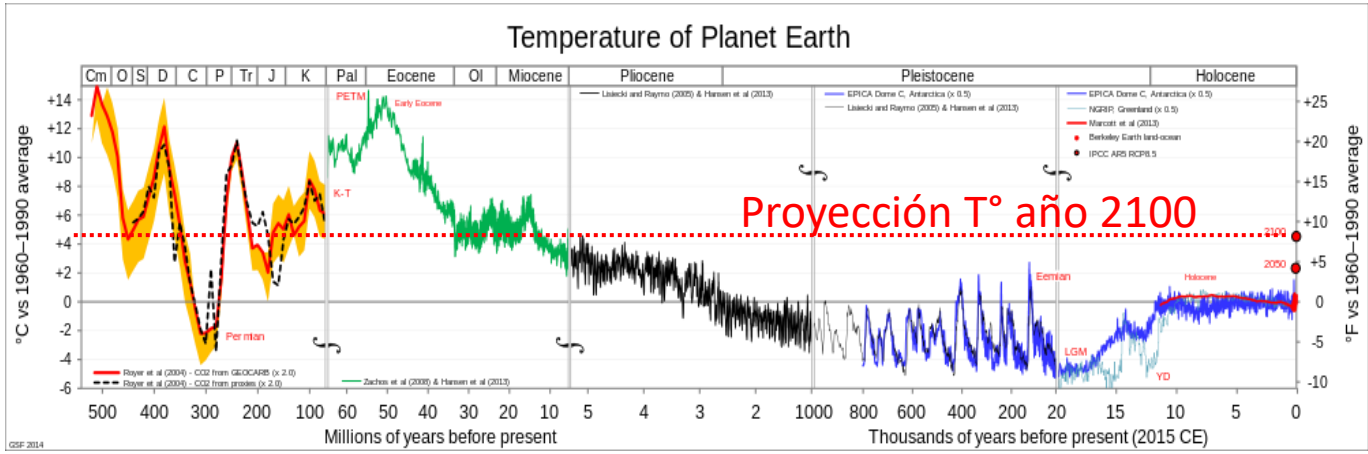
http://www.ces.fau.edu/nasa/impacts

Proyecciones Climáticas

Los **patrones climáticos** son esenciales para determinar el tipo de **ecosistemas** presentes en el planeta, y a su vez, las especies que lo conforman.



Con las proyecciones vigentes, al año 2100, tendríamos las mismas temperaturas que en el Ordovicio, Cretácico y el Oligoceno.



Cambio Global

El **Cambio Global** se refiere a aquellas **transformaciones de gran escala** que están ocurriendo en el **sistema planetario,** que pueden **alterar** significativamente su **funcionamiento y las especies que lo componen,** incluyendo la **capacidad del planeta para sustentar al ser humano**.

Agentes de Cambio

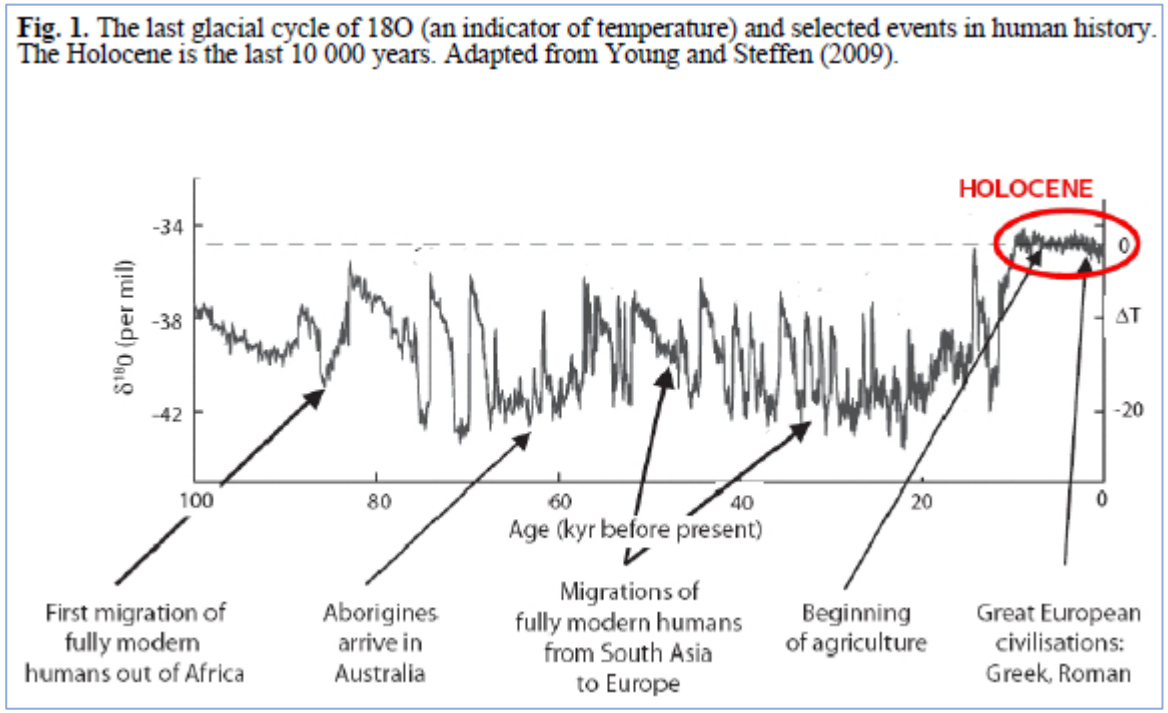
-Crecimiento poblacional
-Sobreexplotación de recursos
-Polución/Contaminación
-Cambio uso de suelo
-Transferencia de biodiversidad

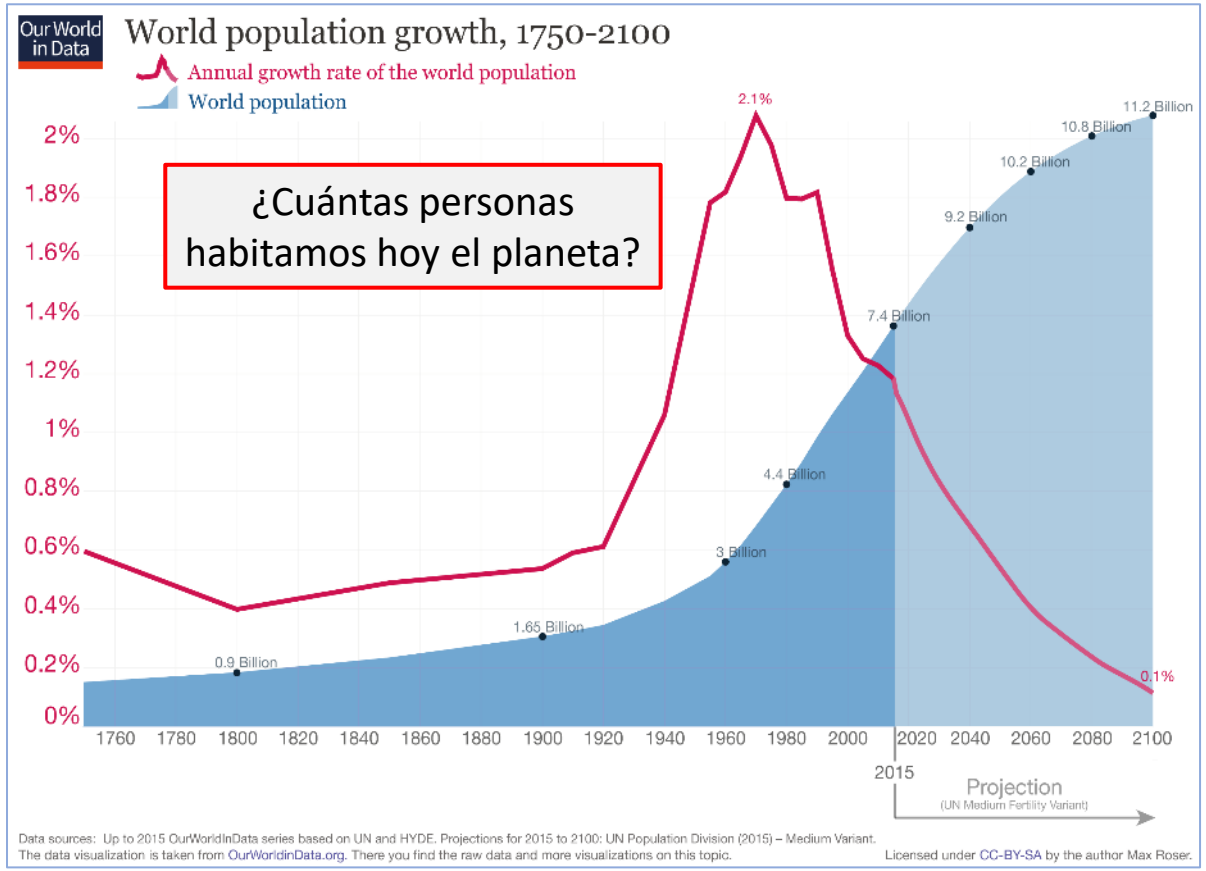
Impactos

-Cambio climático
-Alteración de ciclos biogeoquímicos (H2O, N, P, C, O, S)
-Pérdida de especies
-Conflictos sociales

Agentes de Cambio: Crecimiento Poblacional

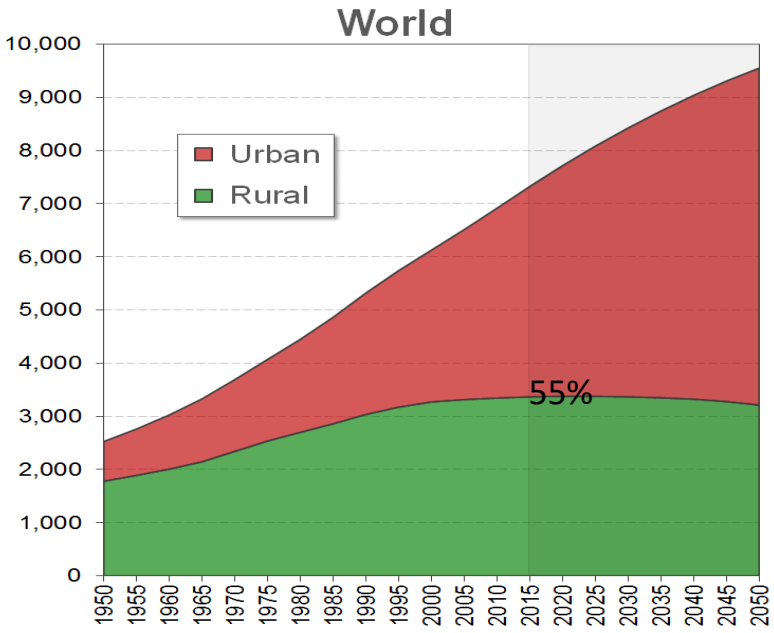
El clima actual ha sido benigno para sustentar a los humanos y sostener su crecimiento poblacional…de manera exponencial en el último siglo.

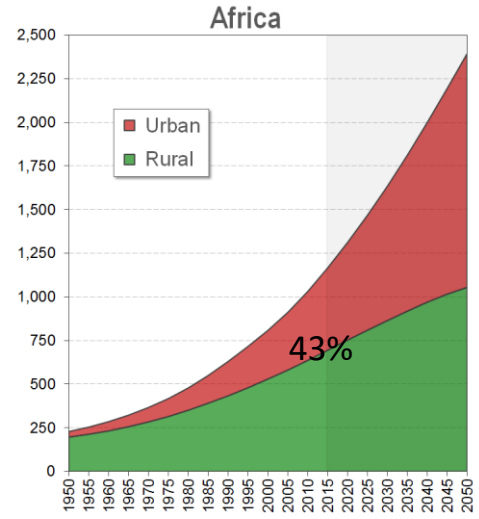


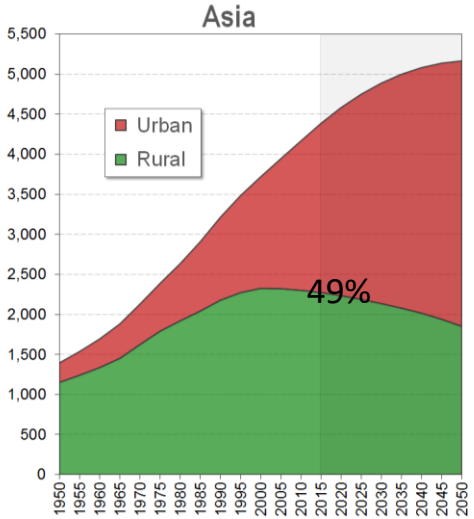


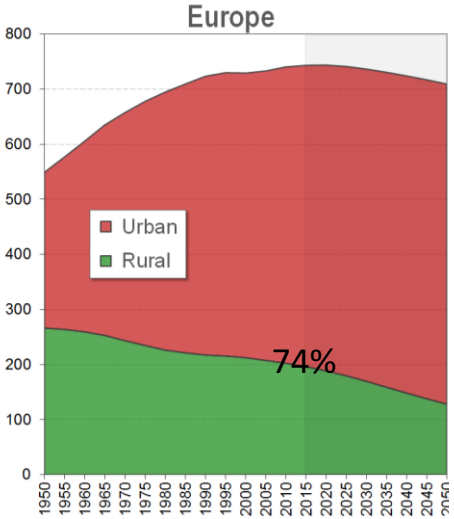
http://www.ecologyandsociety.org/vol14/iss2/art32/

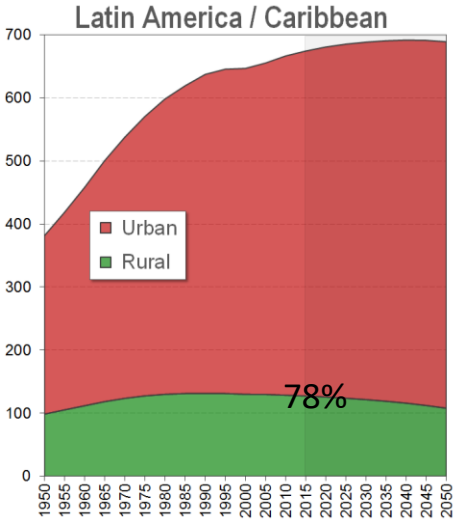
Agentes de Cambio: Crecimiento Poblacional…ciudades

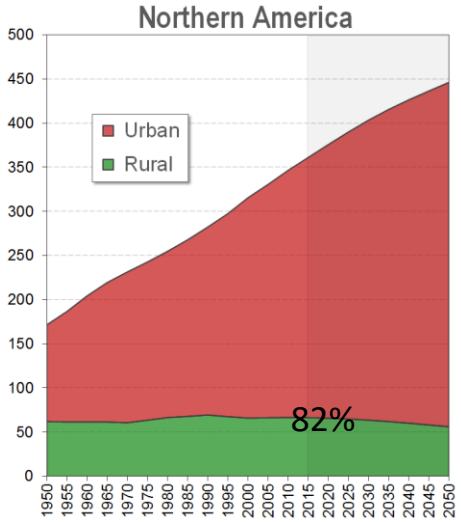








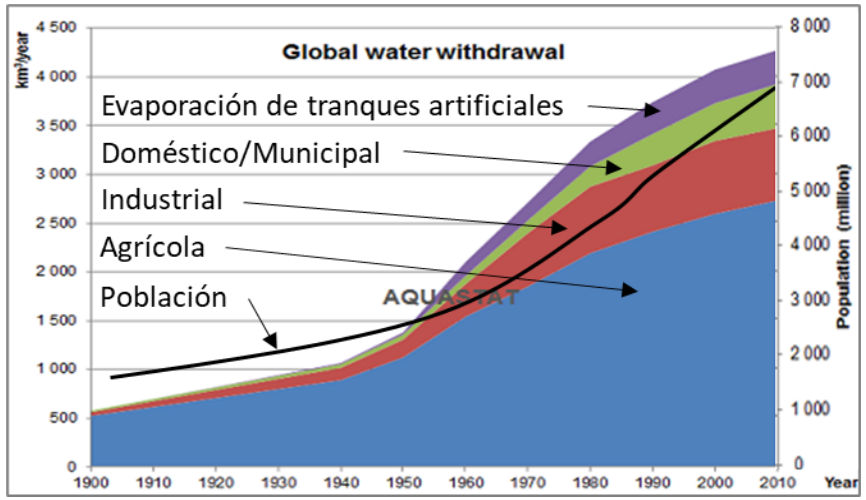


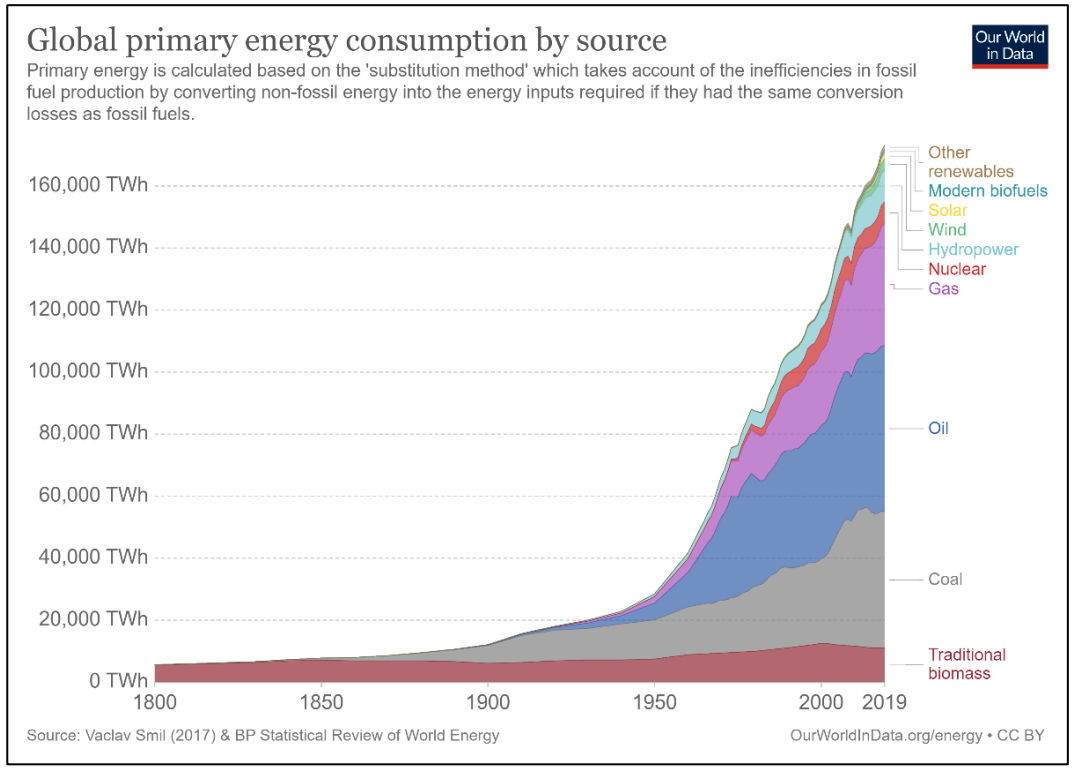


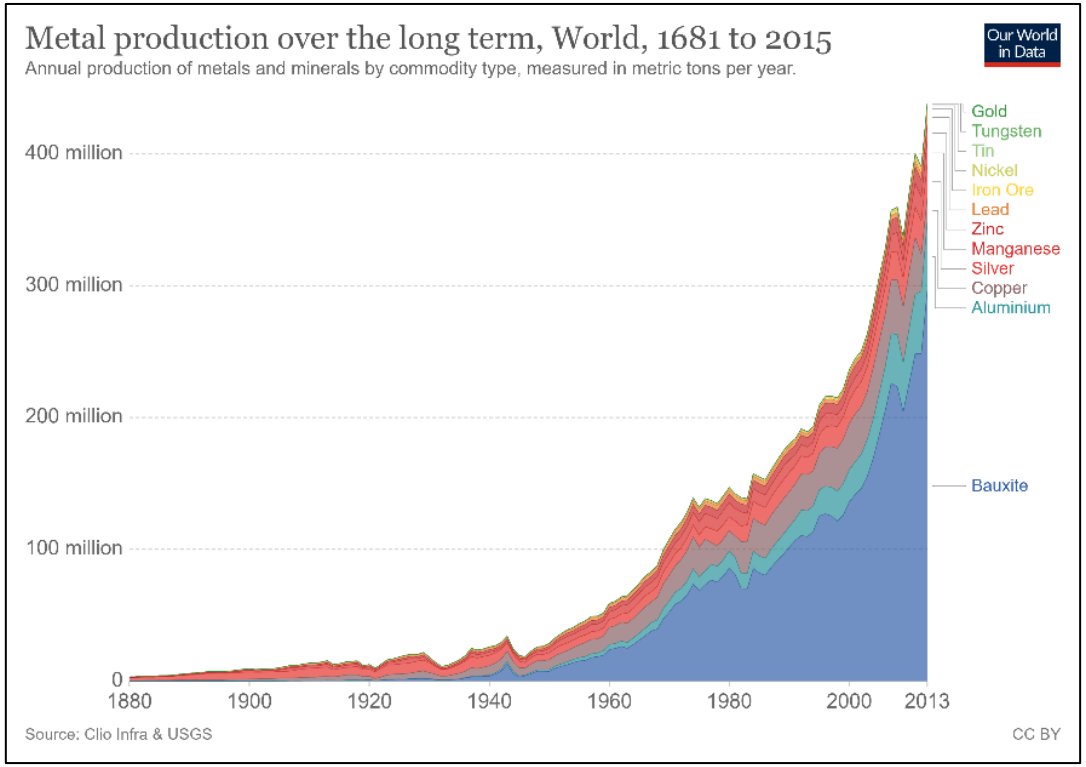
https://ourworldindata.org/

Agentes de Cambio: Sobreexplotación recursos

El aumento poblacional ha implicado un aumento consistente en el uso de recursos naturales como agua, energía y minerales.

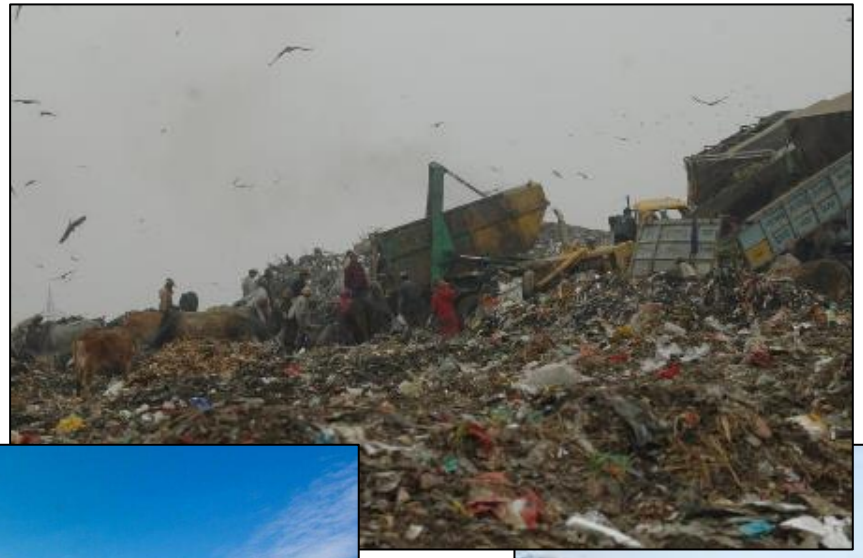




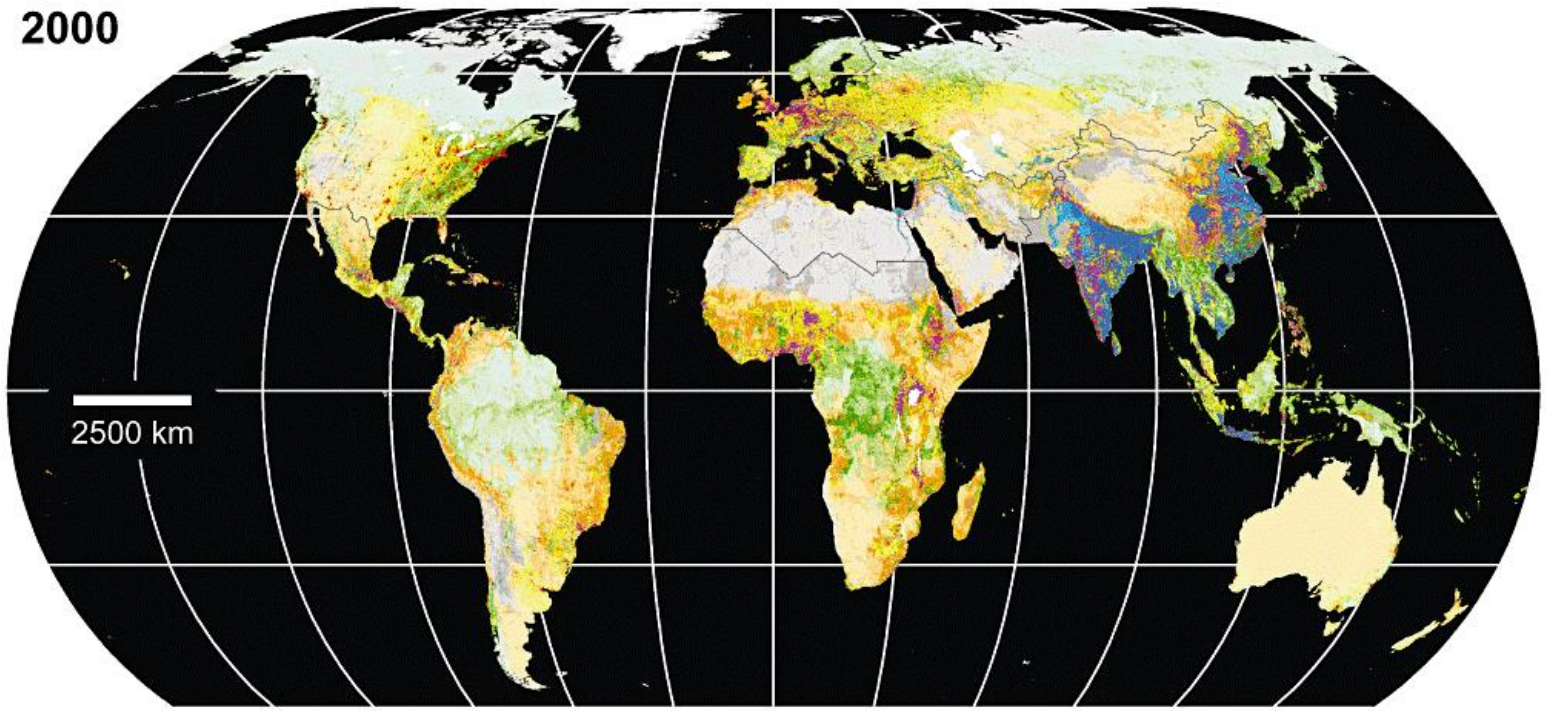


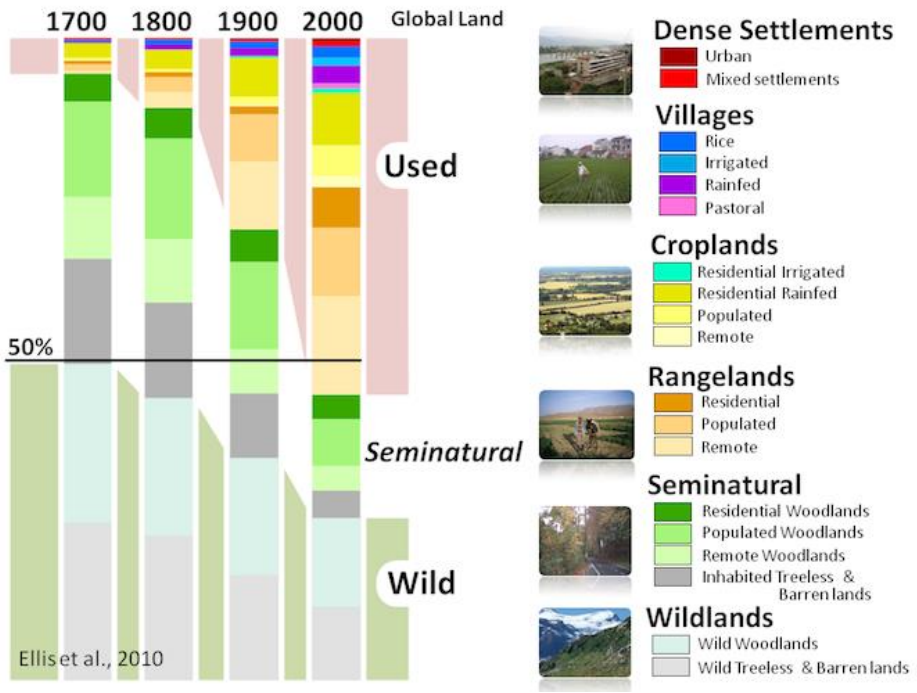
Agentes de Cambio: Contaminación

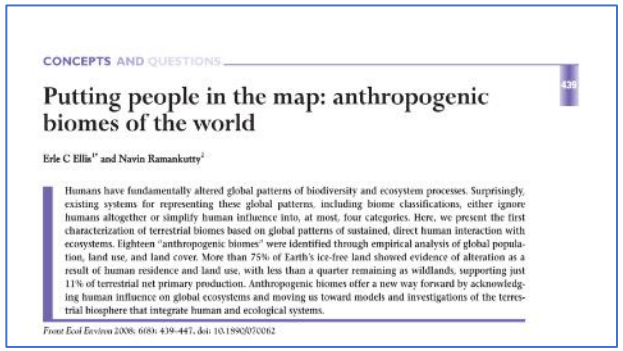
El crecimiento poblacional y “desarrollo” ha traído aparejado la contaminación del planeta a nivel global…agua, tierra, aire.



Agentes de Cambio: Cambio uso de suelo







CONCEPTS AND QUESTIONS

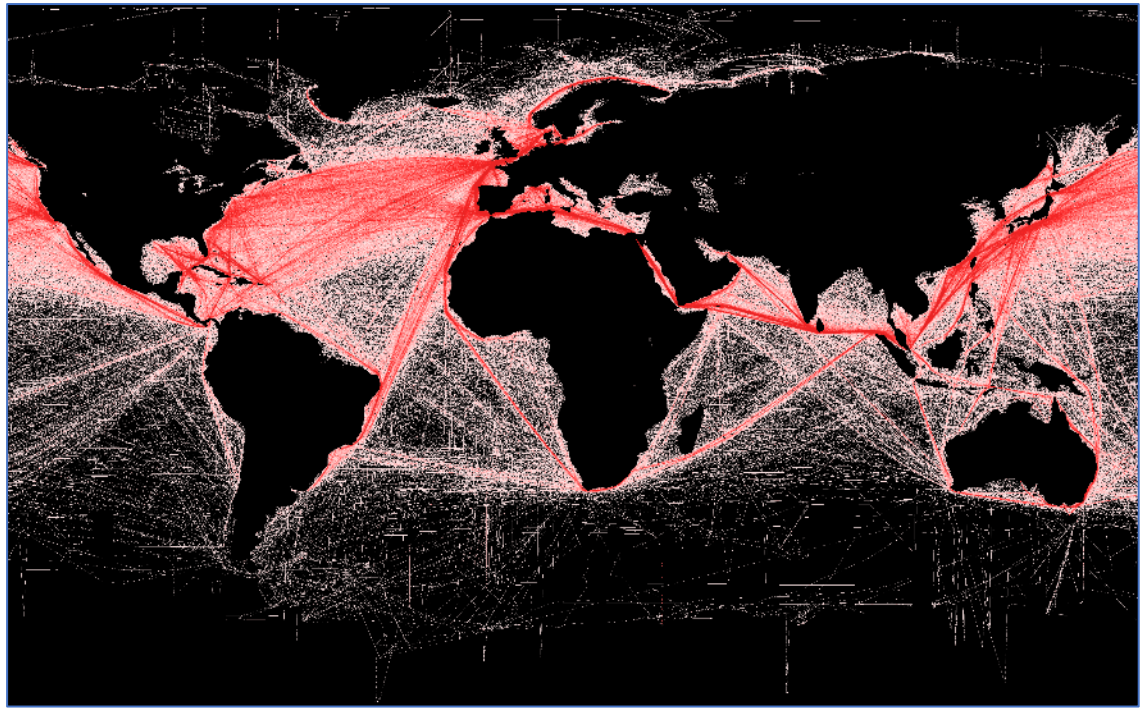
Putting people in the map: anthropogenic biomes of the world

Erle C Ellis[1*] and Navin Ramankutty[2]

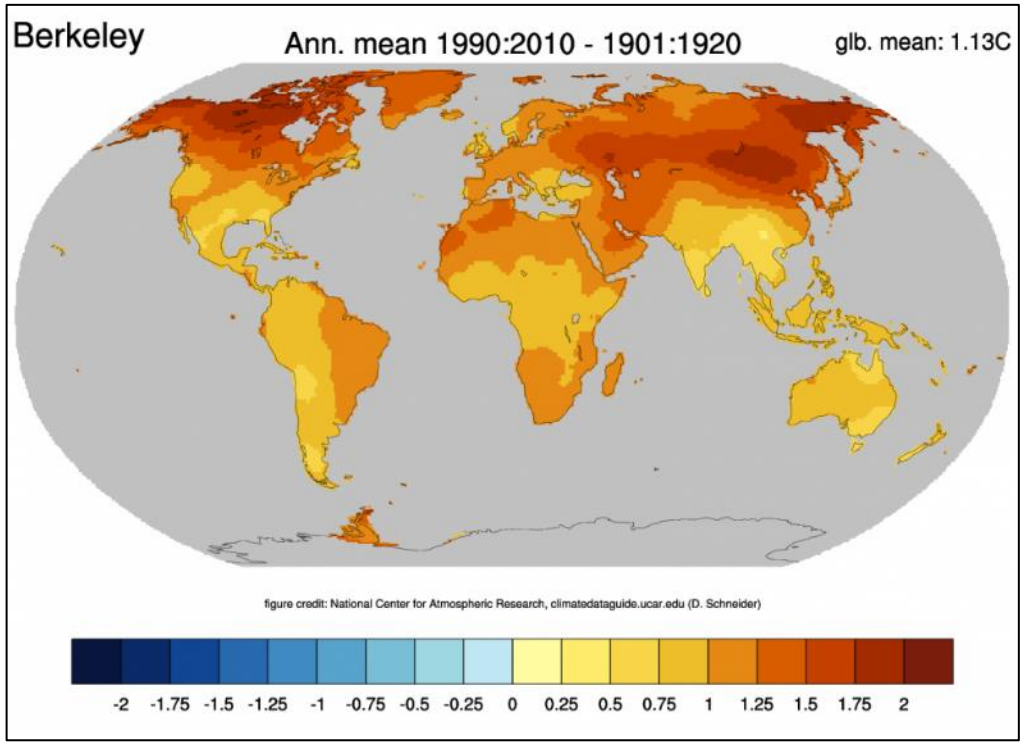
Humans have fundamentally altered global patterns of biodiversity and ecosystem processes. Surprisingly, existing systems for representing these global patterns, including biome classifications, either ignore humans altogether or simplify human influence into, at most, four categories. Here, we present the first characterization of terrestrial biomes based on global patterns of sustained, direct human interaction with ecosystems. Eighteen "anthropogenic biomes" were identified through empirical analysis of global population, land use, and land cover. More than 75% of Earth's ice-free land showed evidence of alteration as a result of human residence and land use, with less than a quarter remaining as wildlands, supporting just 11% of terrestrial net primary production. Anthropogenic biomes offer a new way forward by acknowledging human influence on global ecosystems and moving us toward models and investigations of the terrestrial biosphere that integrate human and ecological systems.

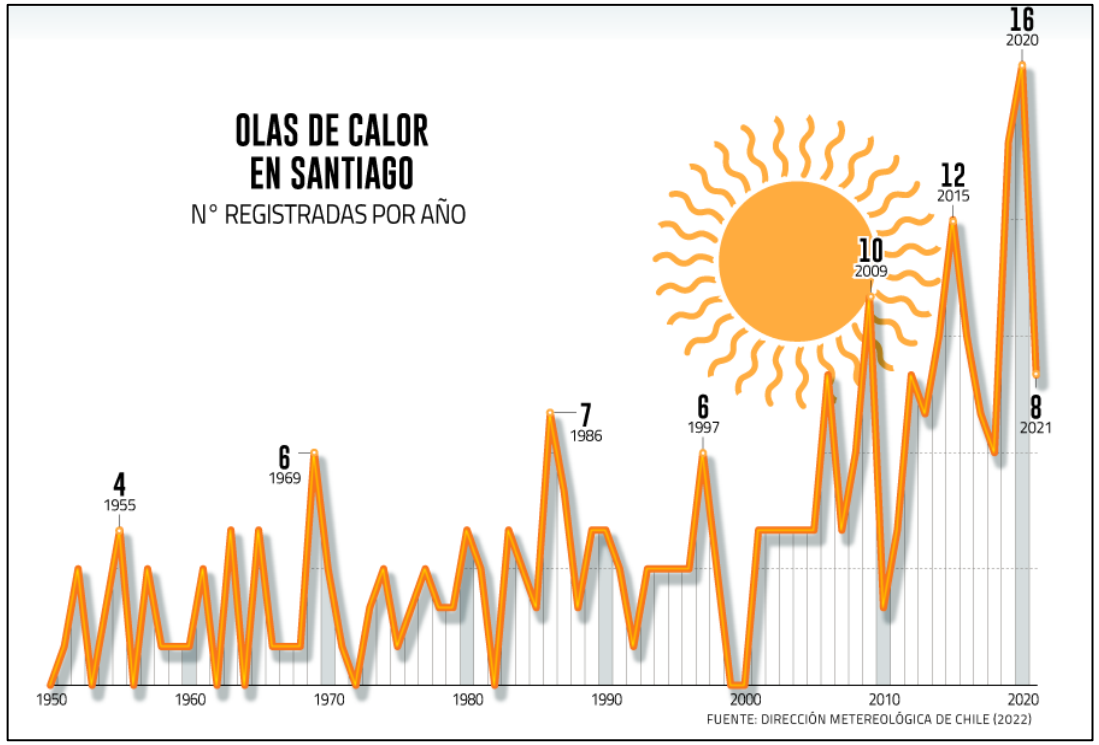
Front Ecol Environ 2008; 6(8): 439–447, doi: 10.1890/070062

Agentes de Cambio: Transferencia biodiversidad



Efectos: Cambio Climático





Efectos: Pérdida de Biodiversidad

Living Planet Index

The Global Biodiversity Outlook 4 (CBD, 2014a) compares the current status of indicators and their projected trends up to 2020 with the UN biodiversity targets. Figure 21 shows what will happen if current trends continue to 2020; by this time vertebrate populations may have declined by an average of 67 per cent since 1970.

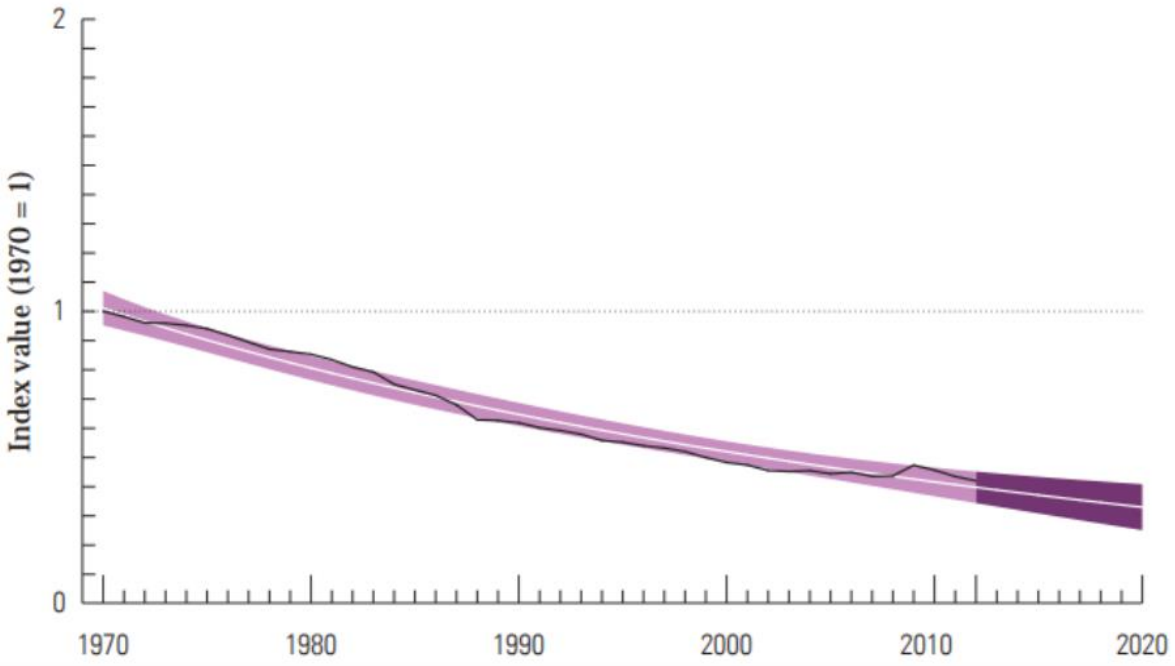
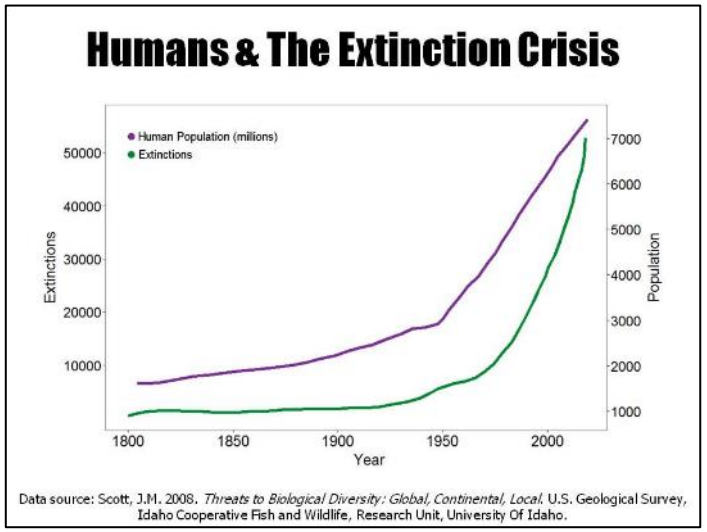
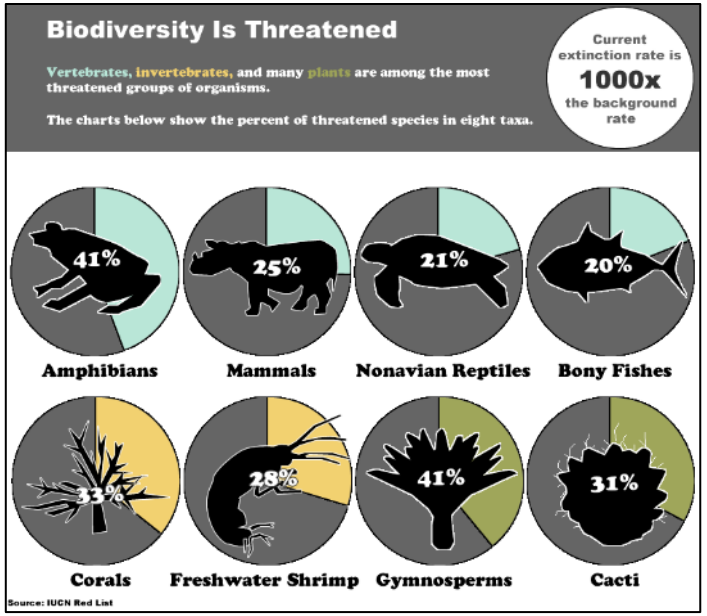


Figure 21: The Living Planet Index extrapolated to 2020 under a business as usual scenario
The Living Planet Index *(solid black line) with model fit and extrapolation to 2020 (white line, shaded area) under a business as usual scenario, shaded band shows 95 per cent confidence limits of the model fit. Using method from Tittensor et al., 2014.*

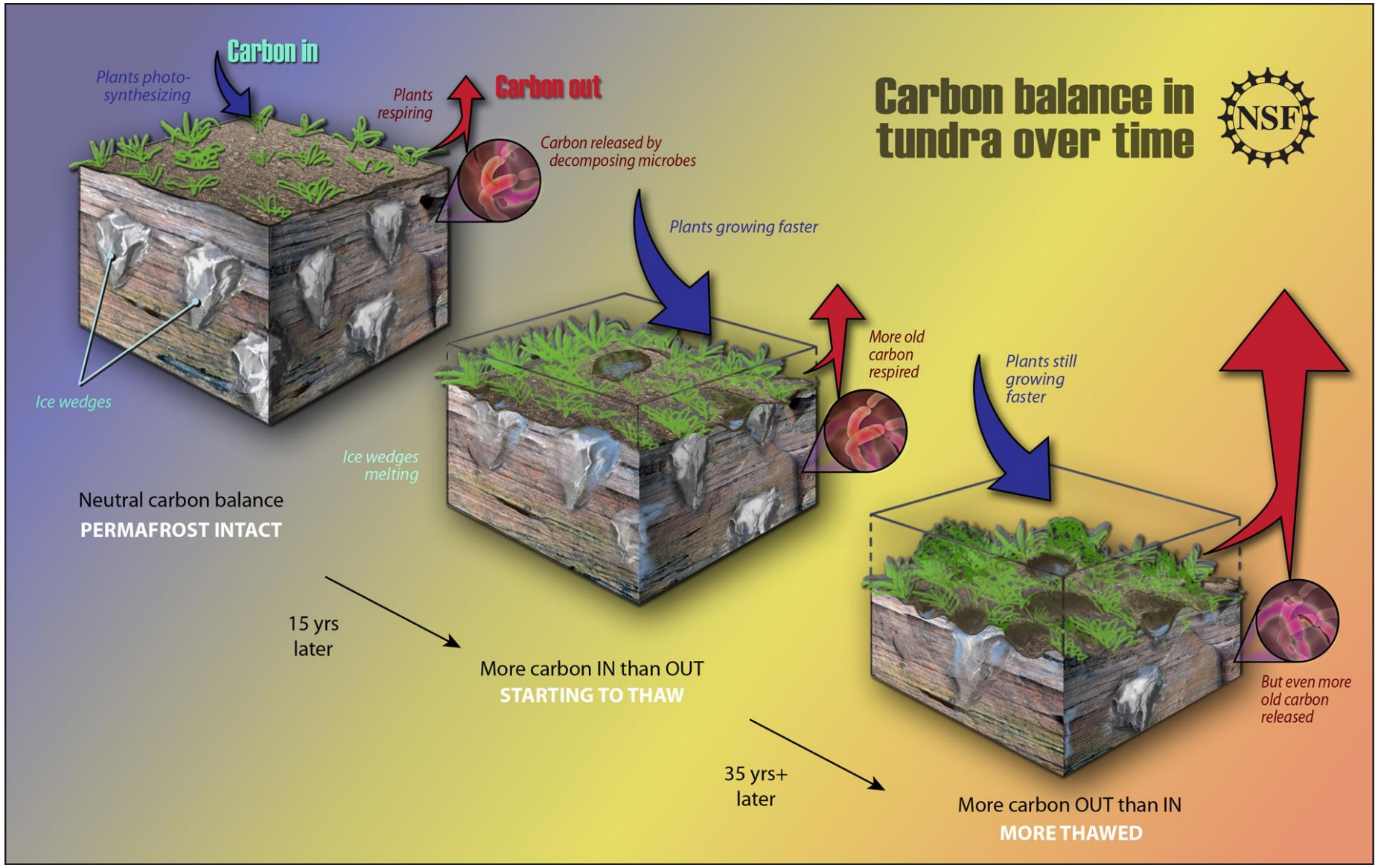
Key

- Global Living Planet Index
- Confidence limits
- Global Living Planet Index - extrapolated
- Confidence limits



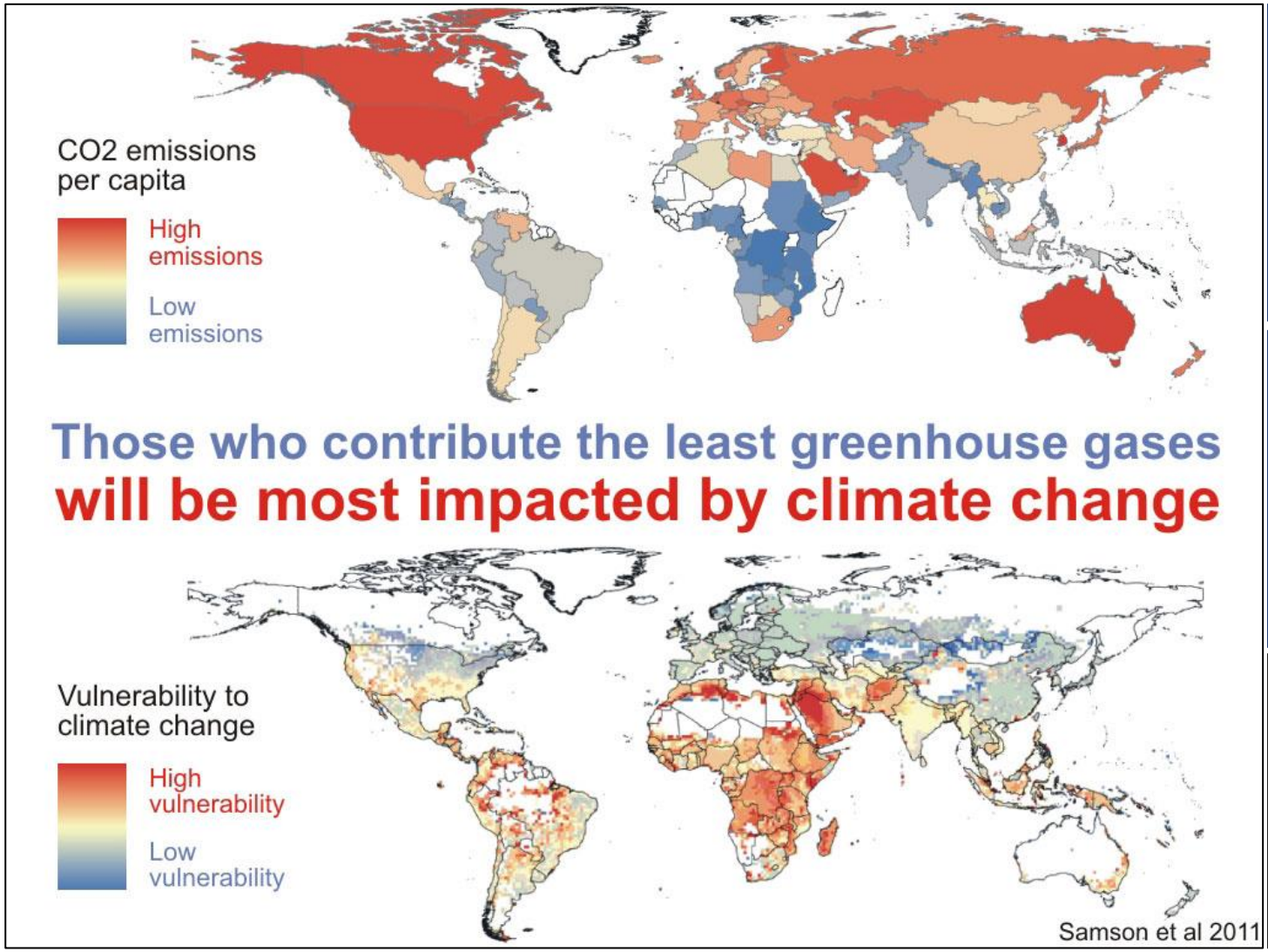


Efectos: Alteración ciclos biogeoquímicos



Efectos: Conflictos Sociales

No sólo afecta la naturaleza, también a los humanos, y de manera injusta.



¿Preguntas, dudas, comentarios?